

2017 计组期中参考答案

一、选择题

BDCDC DDABA BBACC CACBB

二、填空题

1. 0xAD940000 根据要求填空

整型补码	-0x526C0000
无符号整型	0xAD940000
单精度浮点型	$-1.00101_{(2)} \times 2^{-36}$
MIPS 指令	sw \$s4 ,0(\$t4) (或 sw \$20 ,0(\$12))

2. 伪指令转换

伪指令	功能	真实指令
move \$rd, \$rs	\$rd = \$rs	add \$rd, \$rs, \$zero
sne \$rd, \$rs, \$rt	if(\$rs != \$rt) \$rd=1; else \$rd=0;	sub \$at, \$rs, \$rt sltu \$rd, \$zero, \$at
bge \$r1,\$r2,RR	if(\$r1<=\$r2) goto RR	slt \$at, \$r2, \$r1 beq \$at, \$zero, RR
not \$r1, \$r2	\$r1 = ~(\$r2)	nor \$r1, \$r2, \$zero
swap \$r1, \$r2	\$r1 $\longleftrightarrow$ \$r2 仅用两个寄存器	xor \$r1, \$r1, \$r2 xor \$r2, \$r1, \$r2 xor \$r1, \$r1, \$r2

三、MIPS 编程题

1. 设计检测\$s0+\$t0 的 overflow

```
xor $t1, $s0, $t0
slt $t1, $t1, $zero
bne $t1, $zero, Next # 两个操作数异号, 不会 overflow
add $t1, $s0, $t0    # 记录求和结果
xor $t1, $t1, $s0
slt $t1, $t1, $zero # 求和结果与$s0 异号为 1
bne $t1, $zero, Overflow
Next:
```


3. 控制信号设计

RegDst	Jump	Branch	MemRead	MemtoReg	ALUop	MemWrite
X	0	1	0	X	01 ₍₂₎ (-)	0

ALUsrc	RegWrite	bne
0	0	1

附录

eg. 1KB means () bytes.
A: 1 B: 1000 C: 1024 D: 1024*1024

1. MIPS instruction: Add \$s0,\$s1,\$s2, Addressing mode is ().
A: Pseudodirect addressing. B: Register addressing.
C: PC-relative addressing. D: Immediate addressing.

2. According to the IEEE754 single precision, () is $-\infty$.
A: 0xFF00_0000 B: 0xFF10_0000
C: 0xFFFF_FFFF D: 0xFF80_0000

3. IEEE754 single precision representation of -1 is ().
A: 0x3F800000 B: 0xBF100000
C: 0xBF800000 D: 0x3F000000

4. () is not one of Four Design Principles.
A: Good design demands good compromises
B: Simplicity favors regularity

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C: Make the common case fast
D: Smaller is not faster

5. () is a two's complement sign extension of the value 0x7FF0 from 16bits to 32bits?
A: 0x11117FF0 B: 0xFFFF7FF0
C: 0x00007FF0 D: 0x80007FF0

6. 2's complement in 8 bits for -128 is ().
A: Overflow B: 0100 0000
C: 0000 0000 D: 1000 0000

7. () is the hexadecimal equivalent of 105 base 10.
A: 151 B: 54
C: 36 D: 69

8. Suppose 4 8-bit integers (2's complement): r1=0xFE, r2=0xF2, r3=0x90, r4=0xF8. The () will overflow.
A: r2 * r3 B: r1 * r4
C: r2 * r4 D: r1 * r2

9. 0x8000 is a 16-bit biased notation (移碼) representation. the data is ().
A: -32768 B: 0
C: -1 D: +1

10. the pseudo-instruction MOVE:
move rd, rs
the function is rd=rs. Only () cannot do that.
A: and rd, rs, \$zero
B: xor rd, rs, \$zero
C: or rd, rs, \$zero
D: add rd, rs, \$zero

eg.	11	12	13	14	15	16	17	18	19	20
C	B	B	A	C	C	C	A	C	B	B

11. There are no inherent meaning about binary bits:
10101101_00010000_00000000_00000000
The following description which is the most correct ()?
A: A single precision floating-point number: 1.001×2^{-37}
B: It is two's complement of -0x52F00000
C: It represent an integer: -0xAD100000
D: A mips instruction: sw \$t0, 0(\$s0)

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12. Which of the following is not an I-format instruction?
 A: lw \$a0, 5(\$s1)
 B: srl \$a0, \$s1, 5
 C: beq \$a0, \$s1, Five
 D: xori \$a0, \$s1, 5
13. There are two different conventions for ordering the bytes within a word, Little Endian and Big Endian.
 In Little Endian, the byte order for data 0x12345678 in memory is:
 A: 78 56 34 12
 B: 21 43 65 87
 C: 12 34 56 78
 D: 87 65 43 21
14. The abbreviation CRT is the meaning of ().
 A: Color Represent Transfer
 B: Color Ray Television
 C: Cathode Ray Tube
 D: Color Refresh Tool
15. For 2-bit Booth's, the last 3-bit is 110. Y is multiplicand. Then next step will ().
 A: -2Y
 B: +Y
 C: -Y
 D: +2Y
16. The 16-bit biased notation(移码) representation of -1 is ().
 A: 0x8000
 B: 0x8001
 C: 0x7FFF
 D: 0x0001
17. For 8-bit ALU, the () will not overflow(OF=0) but Carry Out(CF=1). integers(2's complement).
 A: 0xFF + 0xFF
 B: 0x7F + 0x80
 C: 0x64 + 0x64
 D: 0x80 + 0x80
18. In the IEEE754 floating point standard, the range of the number depends on (), the precision of the number depends on ().
 A: double precision, normalized number
 B: the fraction, the exponents
 C: the exponents, the fraction
 D: overflow, underflow

19. In MIPS, the register () is saved by the routine making a procedure call. It is called a () register.
 A: \$s, caller-saved
 B: \$t, caller-saved
 C: \$t, callee-saved
 D: \$s, callee-saved
20. () is MIPS instruction "ADD \$t0, \$t1, \$t2" in binary.
 A: 100000 01010 01000 01001 00000 000000
 B: 000000 01001 01010 01000 00000 100000
 C: 000000 01010 01000 01001 00000 100000
 D: 000000 01000 01001 01010 00000 100000

$$60 + 16 + 8 + 3 = 91$$

$$91 - 127 = -36$$

II.1. (10*) The bits have no inherent meaning. Given the number:
 0xAD940000 1010 1101 1001 0100 0000
 What does it represent, assuming that it is:
 0101 0010 0110 1011 1111 1111 1111
 1) a two's complement integer? 128809600 0101 0010 0110 1100 0000 0000 0000

答题情况

二大题

- ori, xori 等是零扩展，不能构造 0xffff ffff
 - 有同学没看清 1-3 条指令实现
 - 有同学没看清只用两个寄存器
 - sw 指令格式，两个寄存器搞反了
 - 用另一条伪指令(e.g. li)表示伪指令
 - 有些同学用一些不常用的指令，sgt, sgtu 没有考虑等号，blez 格式
 - xori 的立即数只有 16 位，没有 32 位
 - beq 后面的 offset 里面所填数字不正确
 - 注意语句执行先后顺序，后执行的会覆盖掉前面的值
- e.g. slt \$rd,\$rs,\$rt
 slt \$rd,\$rt,\$rs
- e.g. bne \$rs,\$rt,next
 add \$rd,\$zero,\$zero
 next: addi \$rd,\$zero,1

三大题

- 第一题只要情况考虑全了都给分
- 第一题有同学不清楚进位和溢出
- 有同学最后没有返回 jr \$ra 都没扣分
- 第二题没注意是 char，应该用字节操作
- 有很多同学用 lb 忽略了它是符号位扩展，这个我没扣分

四大题

- 有同学忘写新加的信号了
- ALUop 是个 2 位控制信号，有些同学与书上有出入，体现减法就算对了